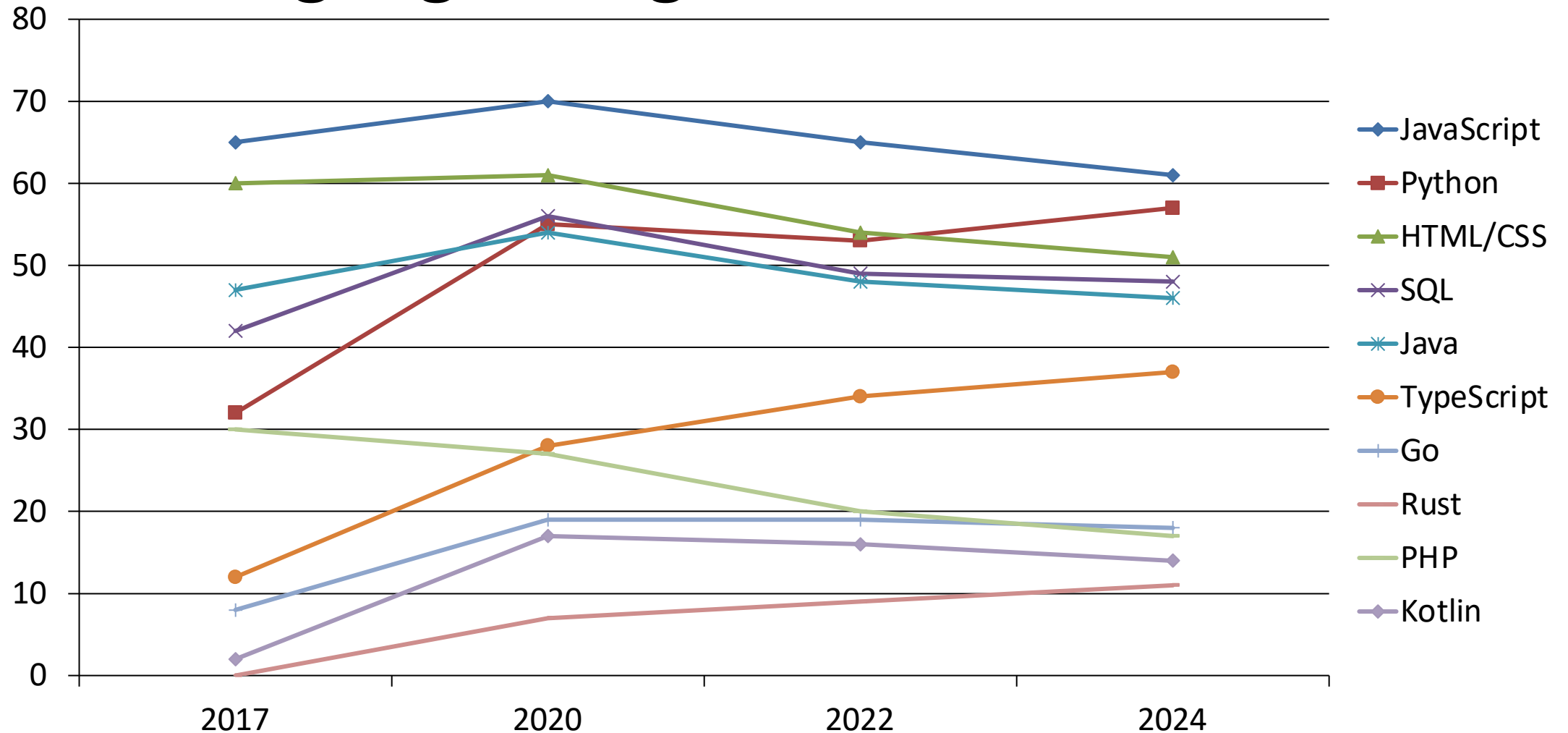


State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

JetBrains' annual survey of 23,262 developers worldwide.

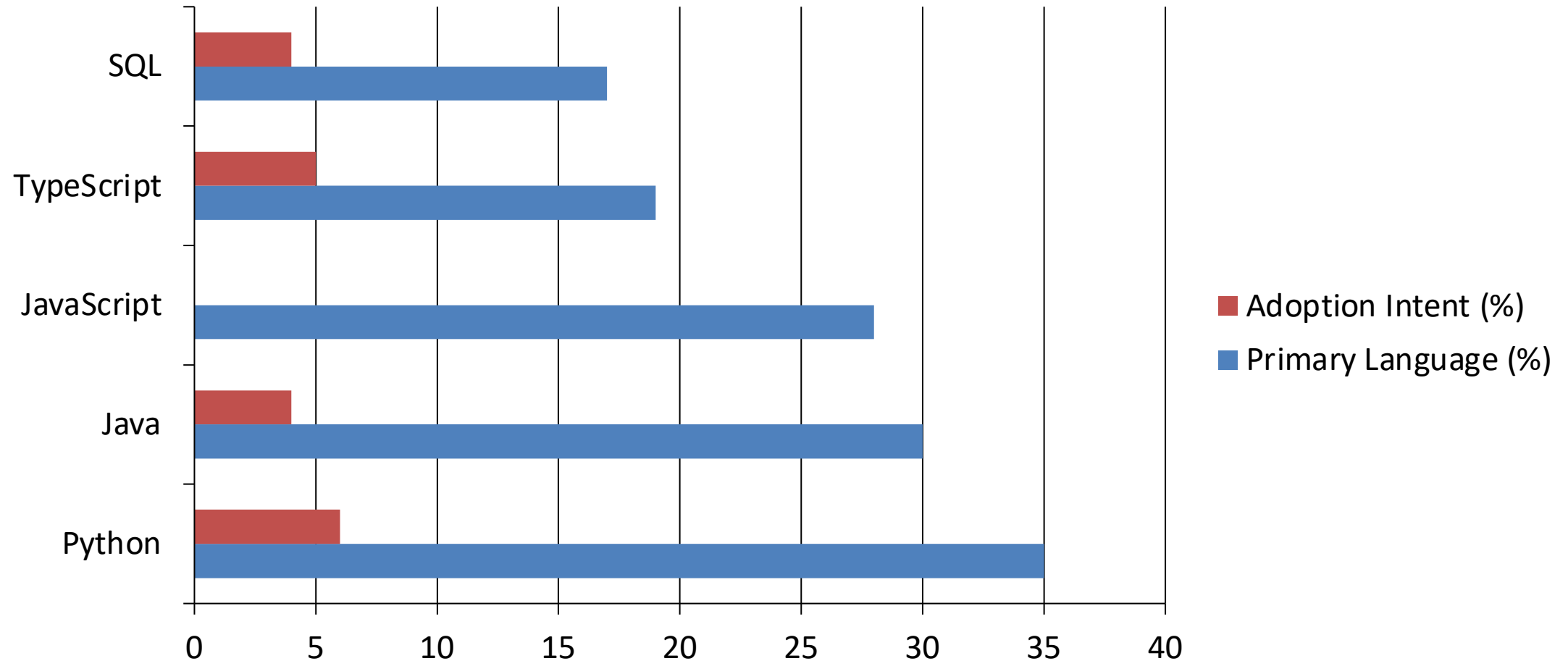
The most comprehensive snapshot of the developer community, covering languages, tools, AI attitudes, and career trends.

Python Closes the Gap: 8 Years of Language Usage Shifts

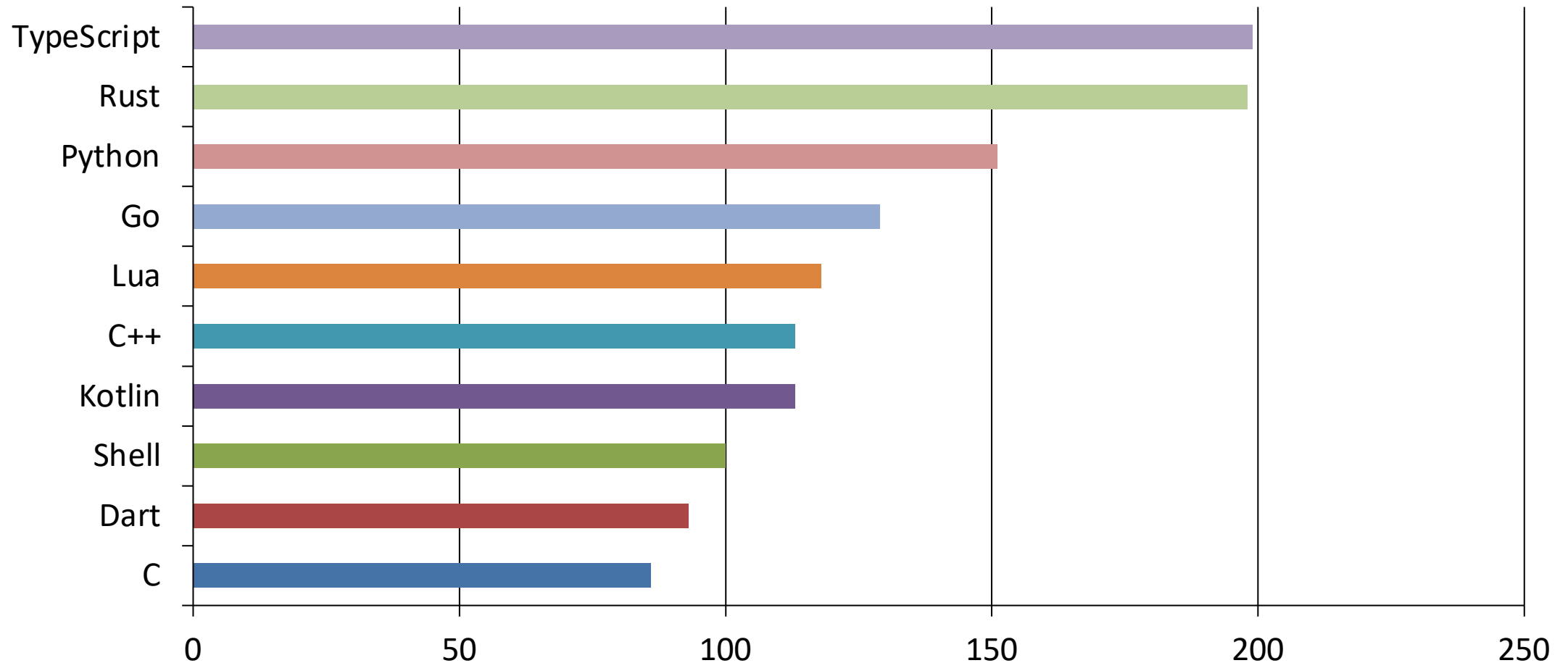


Python Is Now the #1 Primary Language at 35%

Contrast primary usage with adoption intent (all below 6%) to illustrate ecosystem lock-in and the comfort zone effect.



TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198 Satisfaction Score



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Domain	Top Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

What This Means for Your Technology Strategy in 2025

- 1. Invest in Python for AI/ML teams — it is the undisputed leader.
- 2. TypeScript is the future of frontend and full-stack.
- 3. Rust and Go are the growth plays for infrastructure and systems.
- 4. PHP and Ruby hiring pools are shrinking.
- 5. Developer satisfaction correlates with type safety and modern tooling — factor this into retention strategy.